Group Name: Mary and her little lamb's

Section: BSIS – 2A

Brief Description of the System

The Inventory and Sales Management System is a database-driven application designed to help retail businesses track product inventory, record customer sales, and manage supplier information. The system automatically updates stock levels after each sale and provides real-time visibility into product availability and sales performance. It is built to handle a large dataset of over 100,000 transaction records efficiently.

Key Functionalities

Inventory Control: Real-time tracking of stock levels with automated low-stock alerts and SKU-based categorization. Sales & POS Management: Rapid transaction processing with automatic inventory deduction and digital receipt generation. Supplier Management: A registry for contact details and procurement history to streamline restocking. High-Volume Reporting: Advanced filtering and analytics to extract performance insights from large datasets. Security: Role-based access control (Admin, Inventory Clerk, Cashier) to protect data integrity.

Stakeholders

- Store Manager: Analyzes profit, sales trends, and supplier performance.
- Inventory Clerk: Manages stock-in procedures and warehouse audits.
- Cashier: Processes daily customer transactions and checks item availability.
- System Admin: Handles database maintenance and performance tuning.

Core Data Processes (The Continuous Loop)

- Stock Ingestion (Input): Recording new inventory shipments to increment "Stock on Hand" values.
- Transaction Processing (Action): Executing atomic transactions that simultaneously record a sale and decrement inventory to ensure 100% data consistency.
- Data Aggregation (Processing): Indexing the 100k+ record archive to compile instant financial and turnover summaries.
- Restock Trigger (Output): Automated notifications when stock hits pre-defined thresholds, initiating the next procurement cycle.